

Cross-Family Solver Consensus as a Gold-Free Faithfulness Metric for Natural Language to First-Order Logic Translation

Executive Summary

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1 Headline

Cross-family solver consensus—the fraction of independently generated peer translations that are *not* z3-equivalent to a candidate FOL formula—outperforms prompted LLM judges for detecting NL-to-FOL translation errors. On a pre-registered, held-out test on 2,686 real LLM-generated candidates (development dataset E, R_AB labels, 292 sentences), the consensus metric `c_score_align` achieves a stratified AUROC of 0.741, exceeding the best cheap LLM judge (gemini-2.5-flash-lite, disguised) at 0.642: a difference of +0.099 [95% CI: +0.049, +0.146] (Iteration 3, `gen_art_experiment_6`). Consensus matches a frontier judge at roughly 30× lower cost, adds signal beyond a stack of 28 baselines, and requires no gold formulas, no training, and no domain knowledge.

The advantage is confirmed on development data but **not yet on independent confirmation data**: every confirmation arm (E2-A, E2-B, R_COMP FREE) was budget-stopped before reaching testability.

2 Goal

This run investigates gold-free metrics that predict whether a candidate first-order logic formula faithfully captures its source natural-language sentence, without a reference formula, ontology, or domain knowledge. Gold FOL annotations are expensive, non-unique, and unreliable (approximately 42% of entries in FOLIO and MALLS contain incorrect formalisations). The evaluation is a meta-evaluation: we build labelled sets from public benchmarks whose gold has been audited and measure each metric’s ability to discriminate correct translations from errors.

3 What Was Tried

Iteration 1. Screened four metric families head-to-head on a frozen Logic-LM FOLIO screen: FOL-Triage (three-layer cascade), round-trip verification, cross-system consensus, and a disguised cheap LLM judge as the bar. Built development dataset E (700 sentences, 8,507 rows).

Iteration 2. Confirmed the iteration-1 ordering on dataset E; tested PEER+TEXT fusion of consensus and text-based checks; constructed the PERTURB typed perturbation suite (4,234 mutants, 868 controls); re-evaluated all metrics under alternative label regimes.

Iteration 3. Ran the head-on API bar test (T1 CONFIRMED): consensus beats the flash-lite judge by +0.099 AUROC. Tested on R_COMP long composed sentences (SIG within-template

AUROC 0.954 vs judge 0.587). Ran per-operator PERTURB sensitivity and rename tradeoff tests. Verified 48/48 numerical claims from iteration 2.

Iteration 4. Tested Candidate-Signature Consensus (CSC): a provisional negative result due to anchoring ($e = 0.459$ vs 0.173 for free peers). Constructed the E2 confirmation dataset (550 sentences, partial). Produced name-free labels for R_COMP FREE candidates.

Iteration 5. Attempted confirmation on three fronts (E2-A, E2-B, R_COMP FREE); all three budget-stopped before testability. Froze V0 as the shipped metric (no variant V1–V5 met the selection threshold). Tested the Gloss-Gated (GG) name-free checker: failed its development gate. Corrected 15 paper-vs-file mismatches.

4 Key Results

Table 1: Key results across iterations. All numbers from output files.

Result	Value	CI	Source	Grade
c_score AUROC (screen)	0.866	[0.821, 0.906]	Exp 3, Iter 1	DEV
c_score_align strat AUROC (E)	0.741	[0.705, 0.775]	Exp 6, Iter 3	DEV
Δ vs flash-lite disg (strat)	+0.099	[0.049, 0.146]	Exp 6, Iter 3	DEV
Δ vs S4_full (strat)	+0.039	[0.021, 0.057]	Exp 6, Iter 3	DEV
Frontier ratio	0.957	[0.887, 1.034]	Exp 6, Iter 3	DEV
SIG within-template AUROC	0.954	[0.937, 0.965]	Exp 7, Iter 3	OUT-OF-SCOPE
PEER+TEXT pooled AUROC (E)	0.790	[0.752, 0.825]	Exp 5, Iter 2	DEV
k-saturation (k_{95})	3	—	Eval 2, Iter 3	DEV
E2-A DT strat AUROC	0.916	[0.858, 0.970]	Exp 11, Iter 5	CONFIRM

All results are on development data except E2-A DT (confirmation, but only one testable cell of 26 sentences). No confirmatory hypothesis is confirmed or refuted. The full report is 39 pages.

5 What Failed

Table 2: Dead ends and negative results.

Dead end	Iter	Reason
L1 lint on LLM outputs	1	AUROC 0.508 (does not transfer from human errors)
L2 role-aware	1	AUROC 0.557 (below bag-of-words 0.737)
Pilot structural metrics	1	AUROC 0.500–0.533 (null)
Round-trip reformalisation	1	AUROC 0.513 (76% ties)
B2 decomposed z3 reader	2	Gate balanced accuracy 0.476
R_ADJ LLM adjudicator	2	Sonnet-5 balanced accuracy 0.662
Name-free consensus	2–3	SWAP recall 0.512
CSC (candidate-signature)	4	Strat AUROC 0.614; anchoring $e = 0.459$
GG (Gloss-Gated)	5	Dev gate FAIL: strat AUROC 0.696 vs V0 0.742

6 Open Questions

1. **No confirmation.** Every confirmation arm was budget-stopped. The headline +0.099 AU-ROC advantage is on development data only.
2. **Rename tradeoff unsolved.** V0 has RENAME_NONCE false-alarm rate 0.765 on E and 1.0 on E2. No variant achieves both low rename FA and high error recall.
3. **Domain transfer untested.** All evaluation uses FOLIO and MALLS-derived data.
4. **Label noise.** Panel majority accuracy on expert-curated errors is 0.727; R_COMP FREE audit shows 13–28% label contamination.
5. **Complexity advantage inconclusive.** The consensus advantage does not demonstrably grow with sentence complexity (M3 DISCONFIRMED on the length-slope test).